20 There are only red sweets and yellow sweets in a bag.

There are n red sweets in the bag.
There are 8 yellow sweets in the bag.

Sajid is going to take at random a sweet from the bag and eat it.

He says that the probability that the sweet will be red is $\frac{7}{10}$

(a) Show why the probability cannot be $\frac{7}{10}$

P(red) = n/(n + 8). If it were 7/10:

10n = 7(n + 8) = 7n + 56.

3n = 56, so n = 56/3 — not a whole number.

A count of red sweets must be an integer; impossible.

3 marks total
n/(n+8)
10n=7n+56
56/3 not integer

(3)

After Sajid has taken the first sweet from the bag and eaten it, he is going to take at random a second sweet from the bag.

Given that the probability that both the sweets he takes will be red is $\frac{3}{5}$

(b) work out the number of red sweets in the bag.
You must show all your working.

For two reds without replacement:

n/(n + 8) × (n − 1)/(n + 7) = 3/5.

1 mark
Two successive red probabilities

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$5n(n - 1) = 3(n + 8)(n + 7)$.

$5n^2 - 5n = 3n^2 + 45n + 168$.

$2n^2 - 50n - 168 = 0$; *divide by 2:*

$n^2 - 25n - 84 = 0 = (n - 28)(n + 3)$.

$n = 28$ *or* -3; *only 28 is a valid count.*

4 marks total
Form equality 3/5
Quadratic equation
Factorise
n = 28

(5)

(Total for Question 20 is 8 marks)